

# Transfer Speeds Sum

Time limit: 1.00 s

Memory limit: 512 MB

A computer network has  $n$  computers and  $n - 1$  connections between two computers. Information can be exchanged between every pair of computers using the connections.

Each connection has a certain transfer speed. Let  $d(a, b)$  denote the transfer speed between computers  $a$  and  $b$ , which is the speed of the slowest connection on the route between  $a$  and  $b$ . Your task is to compute the sum of transfer speeds between all pairs of computers.

## Input

The first line contains the integer  $n$ : the number of computers. The computers are numbered  $1, 2, \dots, n$ .

After this, there are  $n - 1$  lines, which describe the connections. Each line has three integers  $a, b$  and  $x$ : there is a connection between computers  $a$  and  $b$  with transfer speed  $x$ .

## Output

Print one integer: the sum of transfer speeds.

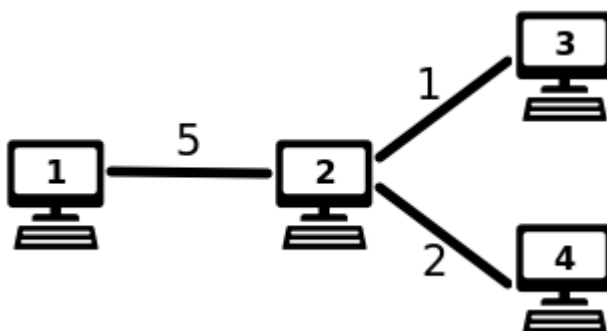
## Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x \leq 10^6$

## Example

| Input                        | Output |
|------------------------------|--------|
| 4<br>1 2 5<br>2 3 1<br>2 4 2 | 12     |

*Explanation:* The following figure corresponds to the sample input:



Here  $d(1, 2) = 5$ ,  $d(1, 3) = 1$ ,  $d(1, 4) = 2$ ,  $d(2, 3) = 1$ ,  $d(2, 4) = 2$ , and  $d(3, 4) = 1$ , so the sum of transfer speeds is 12.